

# Lab Report: 2

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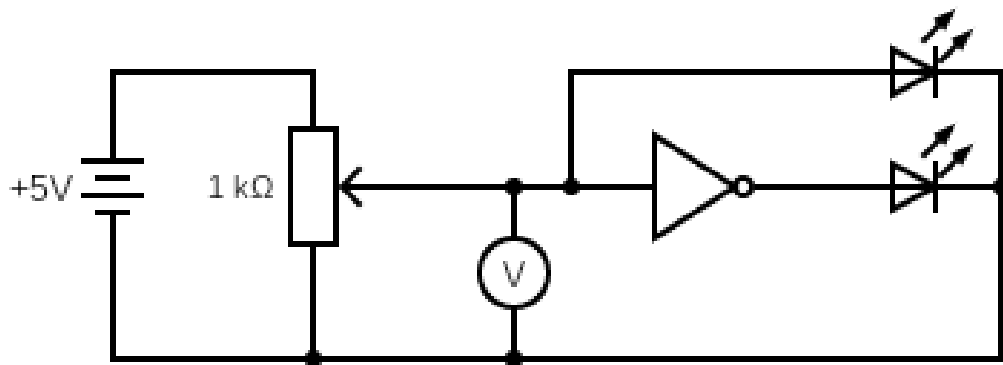
## Part-A

**Objective:** Determining voltage ranges for binary logic levels

**Electronic components used:**

- Digital test kit (w/ breadboard)
- 74HC04 IC
- 1k  $\Omega$  potentiometer
- Multimeter
- Wires

**The reference circuit:**



## Procedure:

- Set up the circuit on the breadboard as shown in the reference diagram.
- Turn the potentiometer shaft to one end; the multimeter reads 0V.
- Rotate the shaft slowly and note the transition of the DP1 LEDs. Note down the voltage range within which the transition occurs till an LED glows completely.
- Now, connect the positive lead of the multimeter to the output pin of the NOT gate and rotate the potentiometer shaft such that the multimeter reading is 0V.
- Again, rotate the shaft while noting down voltage ranges during transition phases.

**Conclusion:** The following came out to be the voltages corresponding to input/output binary logic levels:

- $0 \leq V_{IL} \leq 0.7$
- $2.1 \leq V_{IH} \leq 5.0$
- $V_{OL} \approx 0.19$
- $2.3 \leq V_{OH} \leq 5.0$

Link for the Tinkercad simulation:

<https://www.tinkercad.com/things/3AXPcHDZ7t8-l2parta/editel?sharecode=1-k4dgifRgZOxFzRigczXOSsibTdpG6WFqEbWg2GGzg>

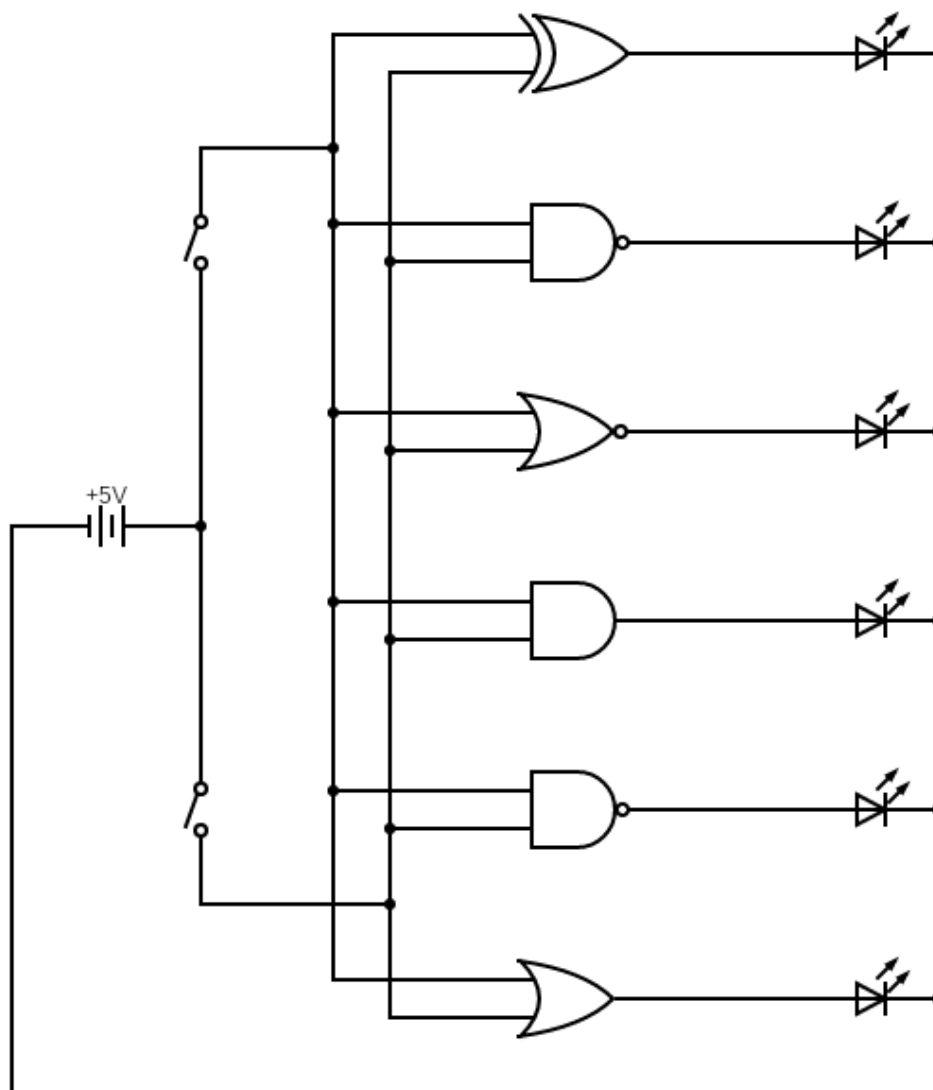
## Part-B

**Objective:** Gate identification via truth tables

**Electronic components used:**

- Digital Test Kit (w/ breadboard)
- Six ICs (NAND/NOR/AND/OR/XOR, TTL LS/CMOS CD)
- Wires

**The reference circuit:**



## Procedure:

- Connect the ICs to the breadboard, power them and make connections as shown in the reference diagram.
- Apply all four possible combinations of inputs to the gates by flipping the switches and tabulate the corresponding outputs of each gate to get the truth table for that gate.
- Identify the gate from the truth table, and look up to verify if the IC corresponds to that gate (from the IC number).
- Additionally, verify the truth table for another gate in the same IC to check if it belongs to the TTL 74LSxx family or CMOS CD40xx family (after looking up the pinout for the same).

**Conclusion:** The six logical gate ICs were identified in the given order:

- 74LS86 (XOR)

| X | Y | $X \oplus Y$ |
|---|---|--------------|
| 0 | 0 | 0            |
| 0 | 1 | 1            |
| 1 | 0 | 1            |
| 1 | 1 | 0            |

- 74LS00 (NAND)

| X | Y | $(X \cdot Y)'$ |
|---|---|----------------|
| 0 | 0 | 1              |
| 0 | 1 | 1              |
| 1 | 0 | 1              |
| 1 | 1 | 0              |

- CD4001 (NOR)

| X | Y | $(X+Y)'$ |
|---|---|----------|
| 0 | 0 | 1        |
| 0 | 1 | 0        |
| 1 | 0 | 0        |
| 1 | 1 | 0        |

- 74LS08 (AND)

| X | Y | $X \cdot Y$ |
|---|---|-------------|
| 0 | 0 | 0           |
| 0 | 1 | 0           |
| 1 | 0 | 0           |
| 1 | 1 | 1           |

- CD4011 (NAND)

| X | Y | $(X \cdot Y)'$ |
|---|---|----------------|
| 0 | 0 | 1              |
| 0 | 1 | 1              |
| 1 | 0 | 1              |
| 1 | 1 | 0              |

- 74LS32 (OR)

| X | Y | $X+Y$ |
|---|---|-------|
| 0 | 0 | 0     |
| 0 | 1 | 1     |
| 1 | 0 | 1     |
| 1 | 1 | 1     |

The NAND gate was repeated, but appeared once in CMOS/TTL configurations, with slight variation in pinout.

Link for the Tinkercad simulation:

[https://www.tinkercad.com/things/71e1KbxU8bZ-l2partb/editel?sharecode=j2TKgchWcaomxfrAQkrzvuwHYtOe\\_M-lEzfjic\\_l8FA](https://www.tinkercad.com/things/71e1KbxU8bZ-l2partb/editel?sharecode=j2TKgchWcaomxfrAQkrzvuwHYtOe_M-lEzfjic_l8FA)

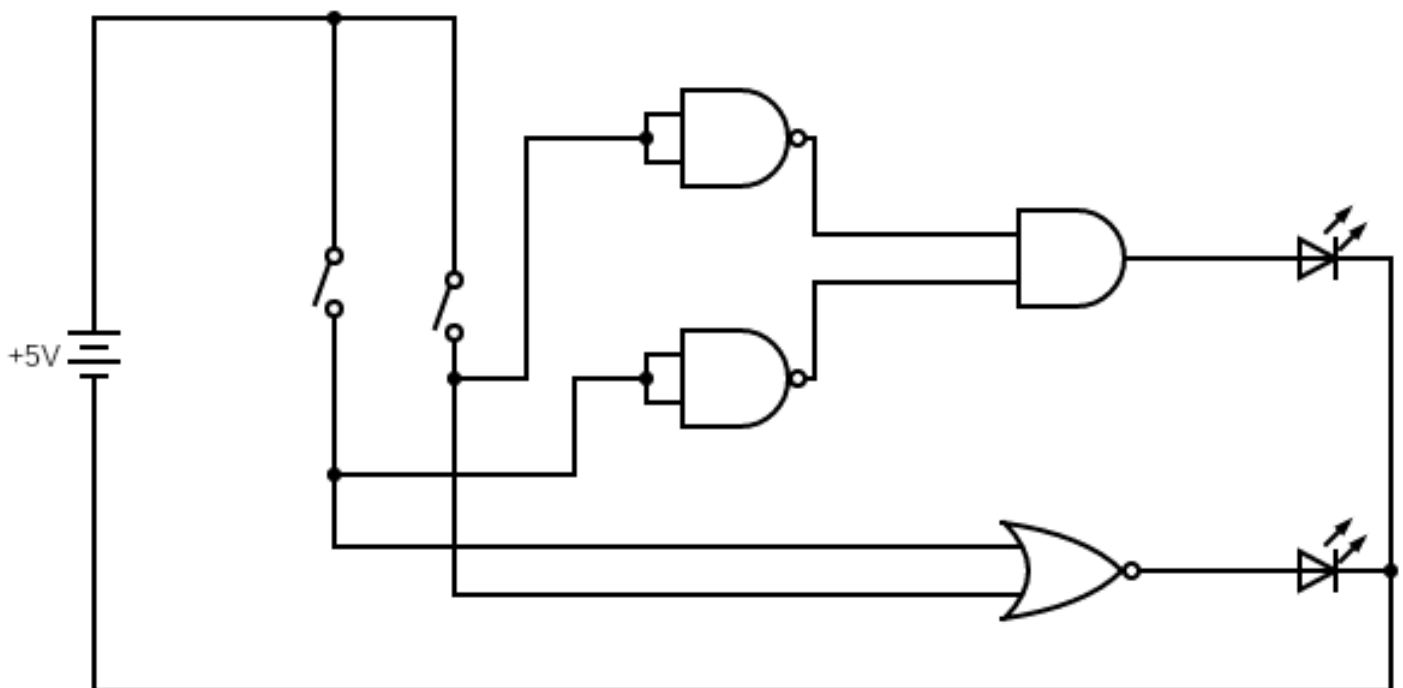
## Part-C

**Objective:** Verification of De Morgan's theorems

**Electronic components used:**

- Digital test kit (w/ breadboard)
- ICs 74LS00, 74LS08 and CD4001
- Wires

**The reference circuit:**



## Procedure:

- Set up the circuit on the breadboard as shown in the reference diagram. Power the ICs and make necessary connections.
- Flip the input switches to get all possible logical input combinations and tabulate the outputs through AND and XOR gates against the inputs.
- Verify if both the truth tables are equivalent, i.e., if De Morgan's law holds (the display LEDs should give same output regardless of input).

**Conclusion:** De Morgan's law –  $(X+Y)' = X' \bullet Y'$  – holds as shown.

| X | Y | X' | Y' | (X+Y)' | X'•Y' |
|---|---|----|----|--------|-------|
| 0 | 0 | 1  | 1  | 1      | 1     |
| 0 | 1 | 1  | 0  | 0      | 0     |
| 1 | 0 | 0  | 1  | 0      | 0     |
| 1 | 1 | 0  | 0  | 0      | 0     |

Link for the Tinkercad simulation:

<https://www.tinkercad.com/things/hWe4E90qNd7-l2partc/editel?sharecode=tGTCQEQAikmBbIFNhMXBTWfRr8oxQAn-z6OcOfIKOE0>

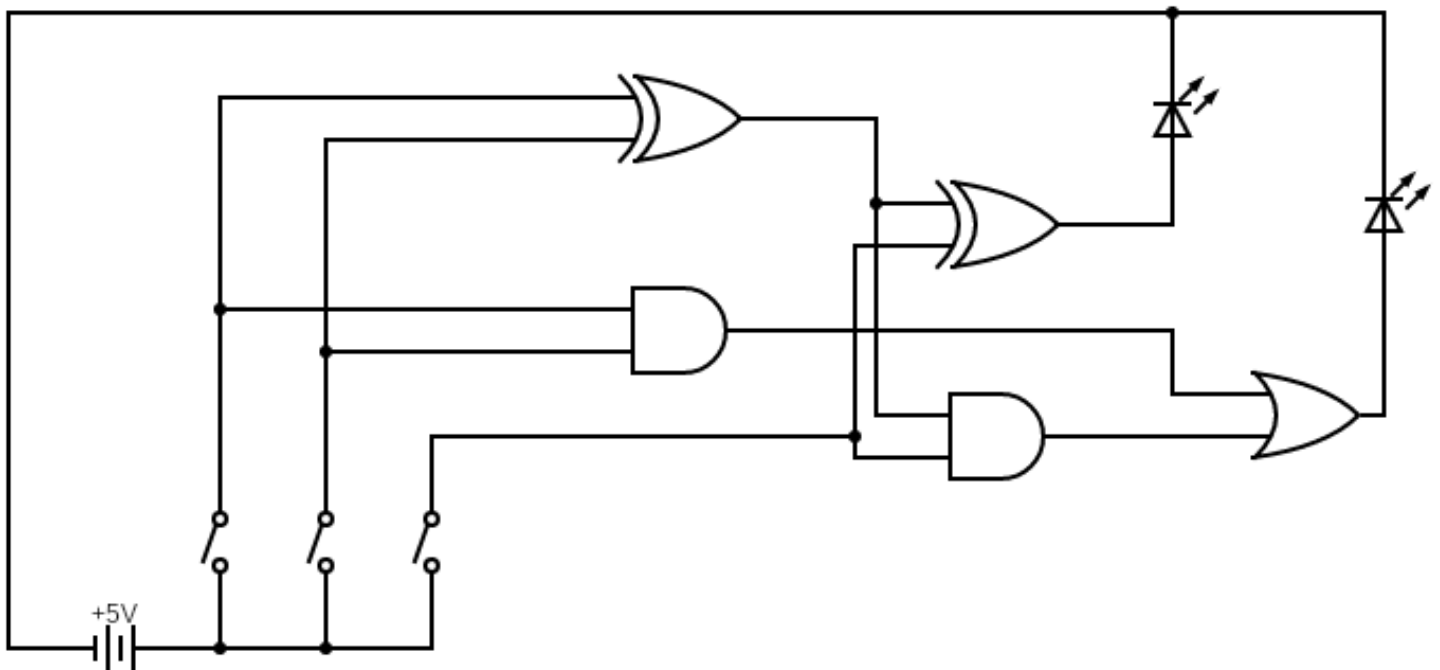
## Part-D

**Objective:** Constructing a binary Full Adder from two Half Adders

**Electronic components used:**

- Digital test kit (w/ breadboard)
- ICs 74LS86, 74LS08 and 74LS32
- Wires

**The reference circuit:**



## Procedure:

- Set up the circuit on the breadboard as shown in the reference diagram. Power the ICs and make necessary connections.
- Obtain the relation between the intermediate carries C1, C2 to get the final CARRY; the gate to be used is OR.
- Observe the SUM and CARRY outputs for all combinations of X, Y and C inputs to the adder. Tabulate the results against the respective inputs into a truth table.

**Conclusion:** The full adder has been constructed with the following results:

| X | Y | C | SUM | CARRY |
|---|---|---|-----|-------|
| 0 | 0 | 0 | 0   | 0     |
| 0 | 0 | 1 | 1   | 0     |
| 0 | 1 | 0 | 1   | 0     |
| 0 | 1 | 1 | 0   | 1     |
| 1 | 0 | 0 | 1   | 0     |
| 1 | 0 | 1 | 0   | 1     |
| 1 | 1 | 0 | 0   | 1     |
| 1 | 1 | 1 | 1   | 1     |

Link for the Tinkercad simulation:

[https://www.tinkercad.com/things/6XVazGxXRd2-l2partd/editel?sharecode=EPES7masconAJjAjqKvdO9R\\_gfb\\_vge4--3\\_mrwiRaI](https://www.tinkercad.com/things/6XVazGxXRd2-l2partd/editel?sharecode=EPES7masconAJjAjqKvdO9R_gfb_vge4--3_mrwiRaI)